Rates of Change & Concavity

ANSWER KEY

Average rate of change

1 Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.
AROC $= \dfrac{f(4) - f(1)}{4 - 1} = \dfrac{24 - 3}{3} = \dfrac{21}{3} = 7$.

2 A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.
$P(0) = 200$. $P(10) = 200(1.05)^{10} \approx 200(1.6289) \approx 325.8$. AROC $= \dfrac{325.8 - 200}{10} \approx 12.6$ people/year.

Increasing/decreasing behavior from rate of change

3 A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.

Since the average rate of change is itself increasing ($2 \to 5 \to 9$), f is speeding up as it increases — this is the signature of a concave-up function on $[0, 3]$.

4 A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.

The average rate of change is decreasing ($8 \to 5 \to 1$), meaning g's growth is slowing down — this indicates g is concave down on $[0, 3]$.

Visualizing concavity and inflection

5 The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.

a Describe the concavity of the graph on $(-2, 0)$. On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.

b Describe the concavity of the graph on $(0, 2)$. On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.

c State what kind of point $x = 0$ represents. $x = 0$ is an inflection point: the location where concavity changes from down to up.

Rate of change of the rate of change

6 Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

A function is concave up on an interval when its average rate of change is itself increasing as you move from left to right across consecutive equal-width sub-intervals — the function isn't just increasing, it's increasing at a faster and faster pace (or decreasing at a slower and slower pace).

Applying rate-of-change reasoning to models

7 A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \le t \le 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.

$R(0) = 0$, $R(2) = 8 - 24 + 80 = 64$, $R(4) = 64 - 96 + 160 = 128$. AROC on $[0, 2]$: $\frac{64 - 0}{2} = 32$. AROC on $[2, 4]$: $\frac{128 - 64}{2} = 32$. Since both average rates are equal, revenue growth appears roughly linear (neither clearly accelerating nor decelerating) over this specific interval pair.

8 For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

$R(6) = 216 - 216 + 240 = 240$. AROC on $[4, 6]$: $\frac{240 - 128}{2} = 56$. Since $56 > 32$ (the AROC on $[2, 4]$), the rate of change is increasing, indicating the revenue curve is concave up on $[4, 6]$ — growth is accelerating in this later interval.

Estimating rate of change from a table

9 A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

AROC on $[0, 1]$: $175 - 120 = 55$ mph. AROC on $[1, 2]$: $238 - 175 = 63$ mph. AROC on $[2, 3]$: $310 - 238 = 72$ mph. Since the AROC is increasing each hour ($55 \to 63 \to 72$), the car's speed is increasing — consistent with a concave-up distance function.

10 Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

AROC on $[0, 3]$: $\frac{310 - 120}{3} = \frac{190}{3} \approx 63.3$ mph. This is an average over the whole trip, blending the slower early hour with the faster later hour — it necessarily falls between the smallest (55) and largest (72) hourly rates, but doesn't equal any single hourly rate.

Visualizing rate of change on a graph

11 The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.

a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$. AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{5} = \frac{3 - 2}{5} = 0.2$.

b Explain how the shrinking steepness in the graph relates to your two AROC values. The curve rises steeply near $x = 0$ and flattens out as x increases, matching the drop from an AROC of 1 down to 0.2 — the function is growing, but at an ever-slower pace.

c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern. Since the AROC is decreasing as x increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

Concavity and real-world interpretation

12 A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

"Increasing but at a decreasing rate" means the function is going up (each AROC is positive) but each successive AROC is smaller than the one before — this is the definition of concave down. So the cost function's graph rises but bends downward, flattening out over time.